

# Critical Dicke electrometry: extending intracavity Rydberg-EIT microwave sensing into the superradiant phase transition

Feasibility study and literature assessment

August 19, 2026

## Abstract

Peng *et al.* [J. Opt. Soc. Am. B **35**, 2272 (2018)] showed that placing a four-level Rydberg-EIT medium inside an optical cavity sharpens the Autler–Townes readout of a microwave electric field. Zhang *et al.* [Phys. Rev. Lett. **110**, 090402 (2013)] showed that Rydberg polaritons in a cavity realise a generalised Dicke model with a superradiant phase transition. This note works out what happens when the two are combined *and the transition itself is used as the sensing element*: the microwave field to be measured sets the dressed-polariton splitting  $\tilde{\omega}_0$ , and therefore moves the Dicke critical coupling  $\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}$ . We construct the effective Hamiltonian, identify three distinct read-out protocols, and evaluate them with exact diagonalisation (closed system,  $N \leq 512$ ) and linearised quantum Langevin theory (open, driven system).

The headline results are mixed, and deliberately so. **(i)** The mechanism is real and the numbers are attractive: the ground-state quantum Fisher information scales as  $N^{4/3}$  at criticality (fitted exponent 1.307), the driven system provides a transduction gain  $\propto \varepsilon^{-1}$  in the distance  $\varepsilon = 1 - \lambda/\lambda_c$  from threshold, up to 10 dB of intrinsic output squeezing, and a quantum-noise-limited sensitivity of  $0.18 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$  at a modest collective excitation  $n_b = 10^4$  — an order of magnitude below the best published Rydberg electrometers. **(ii)** But a control calculation at *fixed* collective excitation shows that the input-referred sensitivity is set entirely by  $n_b$  and the atomic decoherence rate  $\gamma$ , obeying  $\sqrt{S_{\tilde{\omega}_0}} = 0.72\sqrt{\gamma/n_b}$  independently of how close to threshold one sits. Criticality supplies gain and squeezing, not a new scaling law; the  $N^{4/3}$  advantage is a static-protocol artefact that critical slowing down removes, in agreement with the known no-go results. **(iii)** We identify one genuinely novel and defensible contribution: a *null* (threshold-crossing) electrometry protocol in which the measurand is read out as the two-photon laser detuning required to hold the system at threshold, with the same SI traceability as Autler–Townes but a discriminant whose sharpness is set by the transition rather than by the EIT linewidth.

Verdict: worth pursuing, but as a *critical-Dicke superheterodyne amplifier* rather than as a claimed route to super-classical scaling, and the write-up should say so.

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## 1 What is being proposed

The idea, stated compactly: take the intracavity Rydberg-EIT microwave electrometer of Peng *et al.* [1], push it into the regime where the atom–cavity system is described by a Dicke model with a superradiant phase transition (as in the Rydberg-polariton setting of Zhang *et al.* [2]), and use the transition itself — rather than a spectroscopic line splitting — as the microwave-field sensor.

The mechanism that makes this more than an analogy is a simple one. In the four-level ladder  $|g\rangle \rightarrow |e\rangle \rightarrow |r\rangle \rightarrow |r'\rangle$ , the microwave field dresses the two Rydberg states and thereby fixes the transition frequency  $\tilde{\omega}_0$  of the effective two-level system that the cavity mode sees. The Dicke critical coupling depends on exactly that frequency,

$$\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}, \quad \tilde{\omega}_0 = \delta + \Omega_\mu/2, \quad \Omega_\mu = dE_\mu/\hbar, \quad (1)$$

so *the phase boundary is a function of the field to be measured*. Biasing the system just below threshold turns a weak microwave field into a large, directly detected optical signal.

Section 2 builds this Hamiltonian, Sec. 3 lays out three protocols it supports, Sec. 4 evaluates them numerically, Sec. 5 adds the Rydberg interactions that make the link to Ref. [2] explicit, Sec. 6 reviews what has already been done, and Sec. 7 gives an honest assessment.

**A note on sourcing.** The literature survey in Sec. 6 was assembled from search-engine abstracts and metadata; the network policy of the machine this was prepared on blocked direct access to arXiv, APS and Optica full texts. Titles, authors, venues and the quoted headline numbers were each cross-checked against at least two independent search results, but the reasoning inside the cited papers has not been read first-hand. Any claim below about what a cited paper *does not* do should be re-verified before it appears in a manuscript.

## 2 The model

### 2.1 Level scheme and the dressed sensor spin

Take the standard Rydberg-EIT electrometry ladder in  $^{87}\text{Rb}$ :  $|g\rangle = |5S_{1/2}\rangle$ ,  $|e\rangle = |5P_{3/2}\rangle$ ,  $|r\rangle = |53D_{5/2}\rangle$ ,  $|r'\rangle = |54P_{3/2}\rangle$ . A probe field addresses  $|g\rangle \rightarrow |e\rangle$ , a strong control laser  $\Omega_c$  drives  $|e\rangle \rightarrow |r\rangle$ , and the microwave field to be measured drives  $|r\rangle \rightarrow |r'\rangle$  with Rabi frequency  $\Omega_\mu = dE_\mu/\hbar$ . For this transition  $d = 1774.82 ea_0$ , giving the conversion

$$\Omega_\mu/2\pi = 2.271 \text{ MHz} \times (E_\mu/\text{mV cm}^{-1}). \quad (2)$$

On microwave resonance the Rydberg pair is dressed into  $|\pm\rangle = (|r\rangle \pm |r'\rangle)/\sqrt{2}$  at energies  $\pm\Omega_\mu/2$ : this is the Autler–Townes doublet that conventional electrometry reads out as a line splitting. The key move here is to treat one dressed state, say  $|+\rangle$ , together with  $|g\rangle$  as a single effective spin- $\frac{1}{2}$  whose splitting is

$$\tilde{\omega}_0 = \delta + \Omega_\mu/2, \quad (3)$$

$\delta$  being the two-photon detuning of the optical fields. *The measurand has become a Hamiltonian parameter of a two-level system*, not a spectroscopic feature to be resolved.

### 2.2 Engineering the Dicke coupling

A bare dipole coupling cannot produce a superradiant transition: the  $A^2$  term and the Thomas–Reiche–Kuhn sum rule forbid it [15, 16]. The standard way around this is the cavity-assisted Raman scheme of Dimer, Estienne, Parkins and Carmichael [11], realised experimentally by Baden *et al.* [12]: two Raman channels, driven by two control fields with different detunings, generate co-rotating and counter-rotating cavity–atom couplings  $\lambda_{1,2}$  that can be tuned independently. Here this is natural rather than contrived, because the Rydberg-EIT ladder already *is* a Raman configuration; the second control field is the only addition.

The resulting effective Hamiltonian, in the frame rotating at the Raman drive frequencies and after adiabatic elimination of  $|e\rangle$ , is the Dicke model

$$H = \omega a^\dagger a + \tilde{\omega}_0 J_z + \frac{2\lambda}{\sqrt{N}} (a + a^\dagger) J_x, \quad (4)$$

with  $J_\alpha$  the collective spin operators in the  $j = N/2$  manifold. Three features of this construction matter for sensing:

- $\omega$  and  $\tilde{\omega}_0$  are *effective* frequencies — detunings in a rotating frame — and can be dialled anywhere from kHz to MHz. Reaching  $\lambda \sim \sqrt{\omega\tilde{\omega}_0}/2$  therefore does not require ultrastrong coupling.
- $\lambda \propto \Omega_{\text{Raman}} g_0 \sqrt{N}/\Delta_e$  is set by laser power and is tunable *in situ*, so the operating point can be servoed onto threshold.
- The measurand enters through  $\tilde{\omega}_0$  only. That is what makes Eq. (1) a sensing curve.

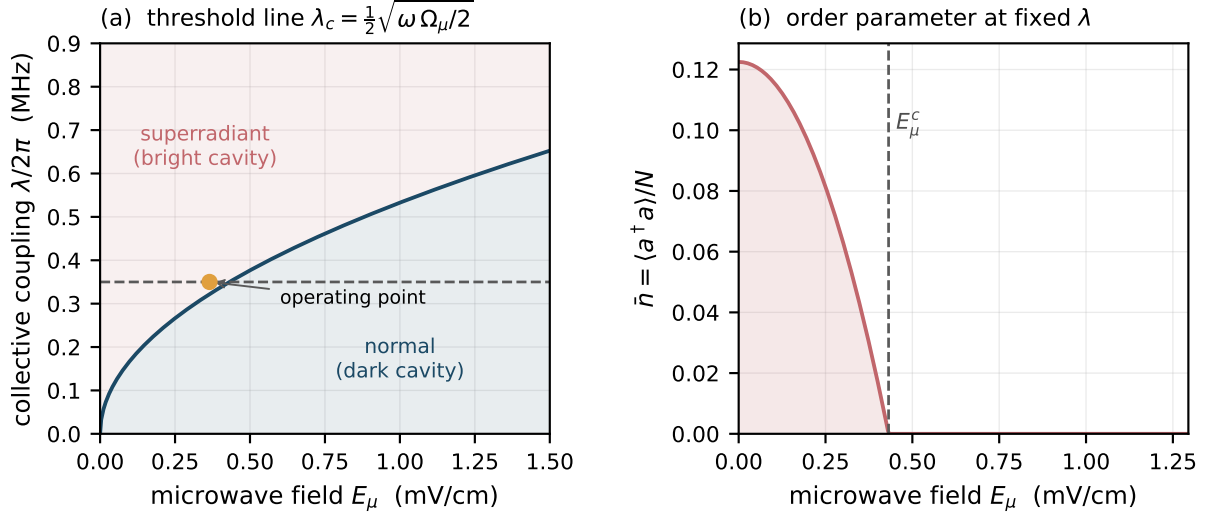
### 2.3 Threshold line and order parameter

Mean-field minimisation of Eq. (4) reproduces the textbook result [14]. Writing  $\langle J_z \rangle = -\frac{N}{2} \cos \theta$ ,  $\langle J_x \rangle = \frac{N}{2} \sin \theta$  and  $\alpha = \langle a \rangle/\sqrt{N}$ ,

$$\frac{E}{N} = \omega \alpha^2 - \frac{\tilde{\omega}_0}{2} \cos \theta + 2\lambda \alpha \sin \theta, \quad \alpha = -\frac{\lambda}{\omega} \sin \theta, \quad (5)$$

which has the normal solution  $\theta = 0$  for  $\lambda < \lambda_c$  and, for  $\lambda > \lambda_c$ ,

$$\cos \theta = \mu \equiv \frac{\lambda_c^2}{\lambda^2} = \frac{\omega \tilde{\omega}_0}{4\lambda^2}, \quad \bar{n} \equiv \frac{\langle a^\dagger a \rangle}{N} = \frac{\lambda^2}{\omega^2} (1 - \mu^2), \quad \lambda_c = \frac{1}{2} \sqrt{\omega \tilde{\omega}_0}. \quad (6)$$



**Figure 1: The microwave field moves the phase boundary.** (a) Threshold coupling  $\lambda_c = \frac{1}{2}\sqrt{\omega\Omega_\mu/2}$  versus microwave field for  $\omega/2\pi = 1$  MHz,  $\delta = 0$  and the  $53D_{5/2} \rightarrow 54P_{3/2}$  dipole. At fixed  $\lambda$  (dashed) the system crosses from superradiant to normal at a well-defined field. (b) Order parameter  $\bar{n}$  at  $\lambda/2\pi = 0.35$  MHz: the cavity goes dark above a critical field  $E_\mu^c$ . The steep square-root onset near  $E_\mu^c$  is the transduction gain the scheme trades on.

Figure 1 shows the resulting sensing curve and the order parameter as the microwave field is swept at fixed  $\lambda$ .

The immediate observation is that the *sign* of the response is the useful one: for  $\delta = 0$  a *larger* microwave field raises  $\tilde{\omega}_0$  and hence  $\lambda_c$ , extinguishing superradiance. The sensor is therefore naturally operated as a bright-to-dark discriminator, which is favourable for detection (one measures a decrease from a large signal, not an increase from zero).

### 3 Three read-out protocols

The construction supports three distinct protocols, with quite different characters. It is worth separating them because most of the confusion in the critical-metrology literature comes from conflating them.

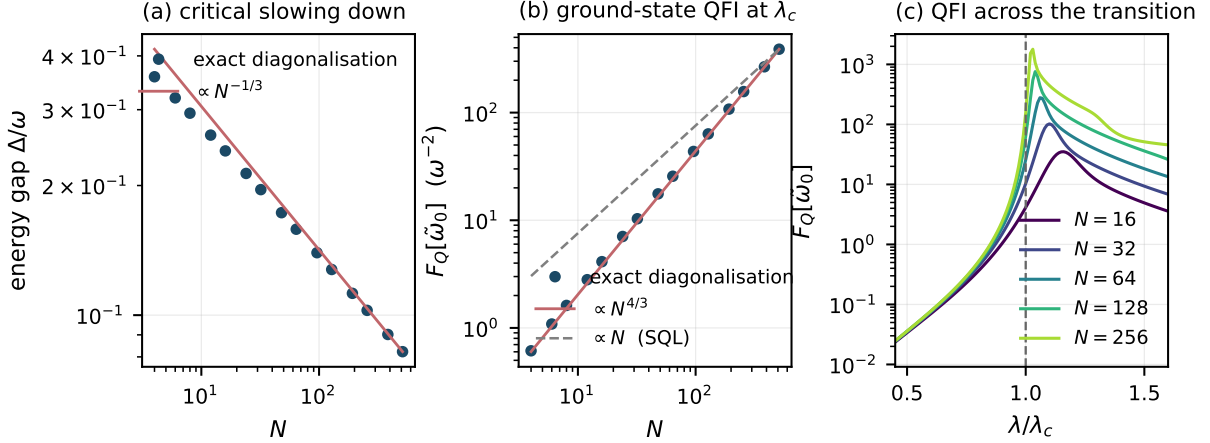
**Protocol A — order-parameter read-out (analogue).** Bias at fixed  $\lambda$  slightly above threshold and monitor the emitted light. The signal is  $\partial\alpha/\partial\tilde{\omega}_0$ , which diverges as  $\mu \rightarrow 1$ :

$$\frac{\partial\alpha}{\partial\tilde{\omega}_0} = \frac{\mu}{4\lambda\sqrt{1-\mu^2}} \xrightarrow{\mu \rightarrow 1} \infty. \quad (7)$$

This is the direct analogue of what Ding *et al.* [6] and Wang *et al.* [7] do with the Rydberg-interaction-driven critical point, and it is the protocol most likely to be implemented first. Its weakness is that the divergent susceptibility is not selective: laser intensity noise (which moves  $\lambda$ ), cavity length drift (which moves  $\omega$ ) and stray dc fields (which move  $\delta$ ) are amplified by exactly the same factor as the signal.

**Protocol B — threshold (null) read-out.** Hold  $\lambda$  and  $\omega$  fixed and servo the two-photon detuning  $\delta$  so that the system sits exactly at threshold. The lock point satisfies

$$\delta_{\text{th}} = \frac{4\lambda^2}{\omega} - \frac{\Omega_\mu}{2} \quad \implies \quad \frac{\partial\delta_{\text{th}}}{\partial E_\mu} = -\frac{d}{2\hbar}, \quad (8)$$



**Figure 2: Finite-size scaling at the critical point** ( $\omega = \tilde{\omega}_0 = 1$ ,  $\lambda = \lambda_c$ ,  $N \leq 512$ ). (a) The energy gap closes as  $N^{-1/3}$  (fitted  $-0.317$  over  $N \geq 128$ ). (b) The quantum Fisher information for  $\tilde{\omega}_0$  grows as  $N^{4/3}$  (fitted  $1.307$  over  $N \geq 128$ ), super-extensive and thus formally beyond the standard quantum limit  $F_Q \propto N$  (dashed). (c)  $F_Q$  across the transition; the peak sharpens and moves towards  $\lambda_c$  as  $N$  grows.

so a change in the microwave field is read out directly as a change in an *optical laser detuning*, which is a frequency and is therefore SI-traceable in exactly the sense that makes Autler–Townes electrometry attractive [9, 10]. The scale factor  $d/2\hbar$  is the same as for Autler–Townes; what changes is the discriminant. In AT electrometry the resolution is set by how well a splitting can be resolved against the EIT linewidth. Here it is set by how sharply the threshold can be located, which is a different — and, near a continuous transition, potentially much steeper — function.

To our reading of the literature this null variant has not been proposed, and it is the most defensible original element of the idea. It also has a practical virtue: null methods are first-order insensitive to gain calibration, which is precisely the weakness of Protocol A.

**Protocol C — critical superheterodyne (recommended).** Apply a microwave local oscillator with Rabi frequency  $\Omega_{\text{LO}}$  to set the bias point  $\tilde{\omega}_0 = \delta + \Omega_{\text{LO}}/2$  just below threshold, and let the weak signal field at  $\omega_{\text{LO}} + \omega_{\text{IF}}$  beat against it:

$$\tilde{\omega}_0(t) = \delta + \frac{\Omega_{\text{LO}}}{2} + \frac{\Omega_{\text{s}}}{2} \cos(\omega_{\text{IF}}t). \quad (9)$$

The measurand is now a small *modulation* of a Hamiltonian parameter, amplified by the near-threshold susceptibility and detected at the intermediate frequency. This inherits the linear-in- $E_{\text{s}}$  scaling and technical-noise rejection that make the Rydberg superheterodyne receiver [8] the most sensitive Rydberg electrometer built, and adds critical gain on top. It is the configuration used for all the open-system numbers in Sec. 4.3, and it is the one we would actually build.

## 4 Metrological analysis

### 4.1 Closed system: quantum Fisher information at criticality

We first ask what the *ground state* of Eq. (4) is worth as a probe of  $\tilde{\omega}_0$ . Exact diagonalisation in the  $j = N/2$  manifold (even-parity sector; photon cutoff converged to  $10^{-12}$ , see Appendix A) gives the quantum Fisher information from the fidelity susceptibility,  $F_Q = 8(1 - \langle \psi(\tilde{\omega}_0) | \psi(\tilde{\omega}_0 + h) \rangle) / h^2$ .

Figure 2 confirms the expected fully-connected-model exponents [18]: the gap closes as  $N^{-1/3}$  and

$$F_Q[\tilde{\omega}_0] \Big|_{\lambda=\lambda_c} \propto N^{4/3}, \quad (10)$$

with a fitted exponent of 1.307 (asymptotic value  $4/3 = 1.333$ ; the residual is the expected slow finite-size correction, the fit drifting monotonically upward as the window moves to larger  $N$ ). In the normal phase we find, at  $N = 512$  and in the window  $N^{-2/3} \ll \varepsilon \ll 1$ ,

$$F_Q[\tilde{\omega}_0] \propto \varepsilon^{-2}, \quad \varepsilon \equiv 1 - \lambda^2/\lambda_c^2, \quad (11)$$

with a fitted exponent of  $-2.0008$  — the Gaussian critical scaling of a squeezed soft mode, cut off at  $\varepsilon_{\min} \sim N^{-2/3}$ , which is exactly what produces the  $N^{4/3}$  peak.

#### 4.2 Why the $N^{4/3}$ is not free

Taken at face value,  $F_Q \propto N^{4/3}$  says  $\delta\Omega_\mu \propto N^{-2/3}$ , beating the  $N^{-1/2}$  standard quantum limit. This is the claim that critical-metrology proposals are often built on, and it does not survive proper resource accounting. The gap closes as  $N^{-1/3}$ , so adiabatic preparation of the critical ground state takes a time  $T_{\text{prep}} \propto N^{1/3}$ . Converting to a per-root-Hertz sensitivity,

$$\delta\Omega_\mu \sqrt{T_{\text{prep}}} \propto N^{-2/3} \cdot N^{1/6} = N^{-1/2}, \quad (12)$$

and the advantage cancels exactly. This is the content of Rams *et al.* [19]; Gietka *et al.* [20] show the cancellation survives shortcuts to adiabaticity, and Gyhm *et al.* [22] have recently established a general  $T^6$  ceiling on the quantum Fisher information of monotonic critical protocols. A proposal built on the  $N^{4/3}$  scaling alone would be building on sand, and this document should not be used to write one.

The interesting question is therefore not the scaling but the *prefactor*, which is what the driven open system delivers.

#### 4.3 Open system: gain, bandwidth, squeezing and the input-referred floor

We linearise Eq. (4) in the normal phase (Holstein–Primakoff bosons  $a$  for the cavity,  $b$  for the collective atomic mode), add cavity decay  $\kappa$  and collective decoherence  $\gamma$ , and drive the cavity coherently. In quadratures  $v = (x_a, p_a, x_b, p_b)$ ,

$$\dot{v} = Av + f + Bu, \quad A = \begin{pmatrix} -\kappa & \omega & 0 & 0 \\ -\omega & -\kappa & -2\lambda & 0 \\ 0 & 0 & -\gamma & \tilde{\omega}_0 \\ -2\lambda & 0 & -\tilde{\omega}_0 & -\gamma \end{pmatrix}, \quad (13)$$

with vacuum inputs  $u$ . The measurand enters as  $\delta H = \delta\tilde{\omega}_0 b^\dagger b$ , i.e. as a coherent force

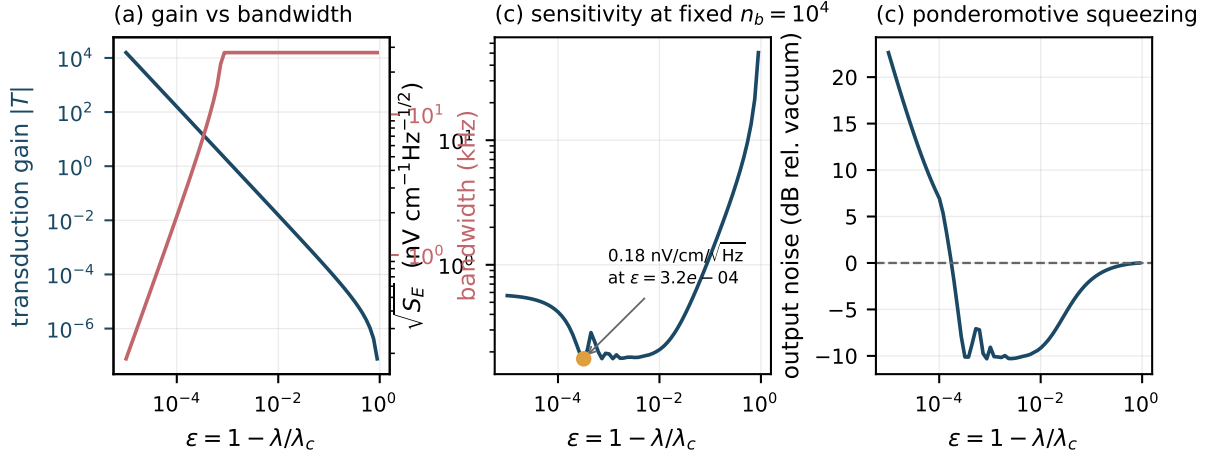
$$s = \delta\tilde{\omega}_0 (0, 0, \bar{p}_b, -\bar{x}_b) \quad (14)$$

proportional to the steady-state collective amplitude — which is why Protocol C needs the probe drive. Output homodyne spectra follow from  $x_{\text{out}} = \sqrt{2\kappa} x_a - x_{\text{in}}$ . The implementation is validated by reproducing the exact vacuum level  $S_{\text{out}} = 1/2$  for an empty cavity.

Throughout we use  $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$  MHz,  $\kappa/2\pi = 50$  kHz and  $\gamma/2\pi = 5$  kHz, giving a damped threshold  $\lambda_c/2\pi = 0.5006$  MHz.

Figure 3 summarises the behaviour. Three points are worth stating precisely.

**Gain and bandwidth.** At fixed collective occupation  $n_b = \langle b^\dagger b \rangle$ , the transduction gain scales as  $\varepsilon^{-1.000}$  over six decades. The bandwidth, however, is *not* immediately paid for: the slowest relaxation rate stays pinned at  $(\kappa + \gamma)/2 = 27.5$  kHz until  $\varepsilon \lesssim 3 \times 10^{-4}$ , below which the soft mode becomes overdamped and the rate falls as  $\varepsilon^{+1.002}$ , giving a strict gain–bandwidth product (constant to 0.78%). Practically: there is roughly three decades of  $\varepsilon$  over which criticality buys  $10^3$  in gain at no bandwidth cost, and beyond that it is a straight trade. This is a genuinely useful design statement and it does not appear to have been made for this system.



**Figure 3: Driven, damped sensor near threshold.** (a) Transduction gain (left axis) and response bandwidth (right axis) versus  $\varepsilon = 1 - \lambda/\lambda_c$ . At fixed collective occupation the gain grows as  $\varepsilon^{-1}$  throughout, while the bandwidth stays pinned at  $(\kappa + \gamma)/2 = 27.5$  kHz until  $\varepsilon \lesssim 3 \times 10^{-4}$  and only then collapses as  $\varepsilon^{+1}$ . There is thus a window of *free* gain before critical slowing down begins. (b) Quantum-noise-limited field sensitivity at fixed  $n_b = 10^4$ , with a broad optimum of  $0.18 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$ . (c) The near-threshold system squeezes its own output by up to 10.3 dB before atomic noise takes over.

**Intrinsic squeezing.** Near threshold the soft mode is a degenerate parametric amplifier and the output is squeezed — we find up to  $-10.3$  dB at  $\varepsilon \approx 9 \times 10^{-4}$ , degrading to excess noise once  $\varepsilon \lesssim 10^{-4}$  where atomic decoherence dominates. The sensor generates its own squeezed light; no external squeezed source is needed. This is a real, if modest, quantum-enhancement channel and it is compatible with Protocol C.

**The control experiment that settles the matter.** Sensitivity figures near a critical point are easy to inflate, because approaching threshold increases the steady-state excitation as well as the gain. We therefore repeated the calculation holding  $n_b$  *fixed* by re-solving for the drive at each  $\varepsilon$ . The result is unambiguous:

$$\boxed{\sqrt{S_{\tilde{\omega}_0}} = 0.72 \sqrt{\gamma/n_b}} \quad (15)$$

with the numerical prefactor constant to one part in  $10^4$  across five decades of  $n_b$  and, away from the extreme critical region, across three decades of  $\varepsilon$ . Equation (15) is the standard quantum limit for  $n_b$  atoms with coherence time  $1/\gamma$ , up to a factor of order unity that we attribute to the intrinsic squeezing.

The physical reading: **criticality is a transducer, not a source of metrological advantage.** It converts the atomic signal into a large optical signal — which is exactly what one needs to beat photodetector and technical noise, and is why Refs. [6, 7] report large Fisher-information gains — but it does not lower the floor set by how many atoms are coherently participating and for how long.

#### 4.4 Sensitivity in physical units

Converting Eq. (15) through  $\delta\tilde{\omega}_0 = \delta\Omega_\mu/2$  and  $E_\mu = \hbar\Omega_\mu/d$ :

Two caveats attach to every number in that table. First, Holstein–Primakoff validity requires  $n_b \ll N$ ; taking  $n_b/N \lesssim 10^{-2}$ , the  $n_b = 10^6$  row needs  $N \gtrsim 10^8$  atoms coherently coupled to one cavity mode, which is demanding but not absurd for a vapour cell and out of reach for a small cold cloud. Second, these are *quantum-noise-limited* figures with no allowance for Doppler broadening, transit-time effects, inhomogeneous coupling across the cavity mode, laser frequency

| Collective occupation $n_b$ | $\sqrt{S_E}$ (nV cm <sup>-1</sup> Hz <sup>-1/2</sup> ) | SQL for $N = n_b$ | ratio |
|-----------------------------|--------------------------------------------------------|-------------------|-------|
| 10 <sup>3</sup>             | 0.567                                                  | 0.786             | 0.72  |
| 10 <sup>4</sup>             | 0.179                                                  | 0.248             | 0.72  |
| 10 <sup>5</sup>             | 0.057                                                  | 0.079             | 0.72  |
| 10 <sup>6</sup>             | 0.018                                                  | 0.025             | 0.72  |
| 10 <sup>7</sup>             | 0.0057                                                 | 0.0079            | 0.72  |

**Table 1:** Quantum-noise-limited field sensitivity at  $\varepsilon = 3 \times 10^{-3}$ ,  $\gamma/2\pi = 5$  kHz. For reference, the Rydberg superheterodyne receiver of Ref. [8] achieves 55 nV cm<sup>-1</sup>Hz<sup>-1/2</sup> and the cavity-enhanced critical measurement of Ref. [7] reports 2.6 nV cm<sup>-1</sup>Hz<sup>-1/2</sup>.

noise or dc-field drift, all of which are what actually limits real Rydberg electrometers. Treat them as an upper bound on what the mechanism can offer, not a prediction.

## 5 Rydberg interactions: the link to the superradiant solid

Reference [2] is not the plain Dicke model: its content is what happens when Rydberg–Rydberg interactions compete with the cavity coupling, producing a superradiant *solid* in which superradiance and crystalline order coexist, with first-order boundaries. It is worth asking whether those interactions help or hurt the sensor.

Adding a nearest-neighbour Rydberg repulsion to Eq. (4),

$$H_{\text{DI}} = H + V \sum_{\langle ij \rangle} n_i n_j, \quad n_i = S_i^z + \frac{1}{2}, \quad (16)$$

and taking a two-sublattice mean field on a bipartite lattice of coordination  $z$  (with  $\alpha$  eliminated by its own stationarity condition),

$$\frac{E}{N} = \omega \alpha^2 - \frac{\tilde{\omega}_0}{4} (\cos \theta_A + \cos \theta_B) + \lambda \alpha (\sin \theta_A + \sin \theta_B) + \frac{Vz}{2} n_A n_B, \quad \alpha = -\frac{\lambda}{2\omega} (\sin \theta_A + \sin \theta_B), \quad (17)$$

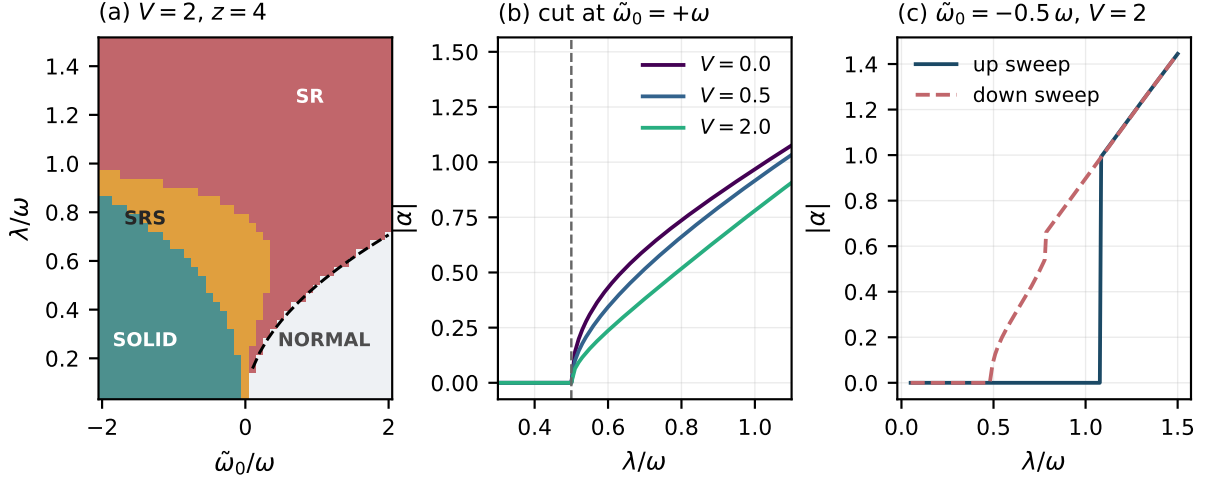
reproduces the phase structure of Ref. [2]. We minimise this from  $13 \times 13$  seeds per point with L-BFGS-B, and use adiabatic up/down sweeps in  $\lambda$  as the first-order diagnostic. Figure 4 shows the result, and it splits cleanly into two regimes.

**At the sensor’s operating point, interactions are inert.** For  $\tilde{\omega}_0 > 0$  the normal-phase Rydberg population vanishes, so  $n_A n_B \rightarrow 0$  and the interaction term does nothing *at* the transition. On a fine scan we find the onset at  $\lambda/\omega = 0.5005$  for every  $V \in \{0, 0.5, 2, 5\}$ , i.e. no shift resolvable at the  $5 \times 10^{-4}$  level. Only the amplitude above threshold is suppressed (Fig. 4b). This is a genuinely useful robustness result: *the calibration curve Eq. (1) is immune to Rydberg interactions*, so atomic density fluctuations — which are exactly what moves the critical point in Refs. [6, 7] — do not move it here. It is the sharpest technical argument for preferring the Dicke critical point over the interaction-driven one.

**The superradiant solid lives elsewhere, and it is first order.** The SOLID and SRS phases appear only for  $\tilde{\omega}_0 < 0$ , i.e. when the dressed Rydberg level sits *below*  $|g\rangle$  in the rotating frame (readily arranged by choosing  $\delta < -\Omega_\mu/2$ ). There the sequence on increasing  $\lambda$  is SOLID  $\rightarrow$  SRS  $\rightarrow$  SR, and the transition is strongly first order: up- and down-sweeps differ by 0.985 in  $|\alpha|$  at  $\tilde{\omega}_0 = -0.5\omega$ ,  $V = 2\omega$ .

The design conclusion, however, is largely independent of the details and is worth stating now, because it runs against the intuition that "more Rydberg physics is better":

- **For Protocols A and C, Rydberg interactions are a nuisance.** The clean collective enhancement of the Dicke model assumes  $N$  independent emitters sharing one bright mode.



**Figure 4: Rydberg interactions and the sensing threshold** ( $V = 2\omega$ ,  $z = 4$ ). (a) Mean-field phase diagram. The superradiant solid (SRS) of Ref. [2] occupies only  $\tilde{\omega}_0 < 0$ ; for  $\tilde{\omega}_0 > 0$  — the sensor’s operating regime — the boundary is the plain Dicke line  $\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}$  (dashed), unmodified. (b) Cuts at  $\tilde{\omega}_0 = +\omega$  for  $V = 0, 0.5, 2$ : the onset does not move, only the post-threshold amplitude is suppressed. (c) At  $\tilde{\omega}_0 = -0.5\omega$  the transition is strongly first order, with a hysteresis loop of height 0.985 in  $|\alpha|$ .

Rydberg blockade caps the number of independently excitable atoms per blockade volume at one, so the effective  $N$  entering  $\lambda = g_0\sqrt{N}$  and the floor (15) is  $N_{\text{eff}} = V_{\text{mode}}/V_{\text{blockade}}$ , not the atom number. Interactions also dephase the collective mode, raising  $\gamma$ . The sensor wants *weak* interactions — moderate  $n$ , or a Rydberg pair chosen for small van der Waals coefficient.

- **For a threshold detector, they are a feature.** If the transition is driven first order, the order parameter jumps discontinuously and the system latches, with hysteresis. That is useless for analogue metrology but excellent for a *trigger*: a device that fires when the field crosses a settable level, with essentially unbounded gain at the switching point and built-in hysteresis to suppress chatter. This is a legitimate secondary application and it is the one place where Ref. [2]’s first-order physics is directly useful.

The recent proposal of an anisotropic Dicke–Ising model in driven Rydberg arrays [28] covers the equilibrium phase structure of this regime; it does not address sensing.

## 6 Prior art

The relevant literature falls into four strands. The purpose of this section is to be honest about how much of the idea is already occupied.

### 6.1 Cavity-enhanced Rydberg electrometry — the direct lineage

Peng and co-workers have published a coherent programme along exactly this line. Reference [1], the seed paper, places a four-level Rydberg-EIT medium in a cavity and reports a minimum detectable field roughly  $8\times$  smaller than the cavity-free case, attributing the gain to cavity narrowing of the transmission features. The same group then moved to the collective regime: Ref. [3] uses *collective Rabi splitting* — i.e.  $g_0\sqrt{N}$  — and reports about  $7\times$  improvement over plain EIT, and Ref. [4] reports  $196.7\times$  over a bare atomic medium and  $26.2\times$  over weak coupling. Experimentally, Ref. [5] demonstrates 18 dB of power-sensitivity enhancement using a microwave (not optical) resonator.

**Implication for the idea.** The collective-coupling step is already taken. Anything proposed here must be distinguished from “ $g_0\sqrt{N}$  in a cavity”, because Ref. [3] is precisely that. The

distinguishing feature has to be the *counter-rotating* terms and the phase transition they enable — Refs. [3, 4] are Tavis–Cummings physics in the normal phase, with no transition and no critical point.

## 6.2 Critical Rydberg metrology — the closest competitor

Ding, Liu, Shi, Guo, Mølmer and Adams [6] demonstrated many-body enhanced microwave metrology at the critical point of a non-equilibrium Rydberg gas, reporting a Fisher information three orders of magnitude above the independent-particle value and an equivalent sensitivity of  $49 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$ . Wang, Liang, Wang and co-workers [7] then put that critical system inside an optical cavity, reporting more than two orders of magnitude of additional Fisher information and  $2.6 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$ .

**This is the strongest prior art and it must be addressed head-on.** Reference [7] is “critical Rydberg metrology in a cavity”, which is a fair one-line summary of the present idea too. The substantive difference is *which* criticality:

- Refs. [6, 7] use the *dissipative, Rydberg-interaction-driven* phase transition (optical bistability of the driven Rydberg ensemble). Its critical point is set by atomic density and interaction strength, and therefore drifts with temperature, density and stray fields, and cannot be tuned quickly.
- The present proposal uses the *light-matter* (Dicke) transition. Its critical point is set by  $\lambda \propto$  laser amplitude and by  $\omega$ , both of which are tunable in situ on microsecond timescales and can be servoed. The critical point is a controlled instrument setting rather than a property of the sample.

That difference is real and is worth a paper, but it is a difference of *controllability and traceability*, not of sensitivity scaling. Claiming the latter would be wrong (see Sec. 4.3).

## 6.3 Critical quantum metrology — the theory that constrains the claims

Garbe *et al.* [17] established the finite-component critical metrology paradigm with the quantum Rabi model, and Ref. [18] worked out the fully-connected-model scaling (static, quench and finite-time-ramp protocols, including Kibble–Zurek scaling) whose exponents we reproduce in Fig. 2. Against these, Rams *et al.* [19], Gietka *et al.* [20, 21] and Gyhm *et al.* [22] constrain what can be claimed once time is counted as a resource. Any write-up of this idea must engage with that second group of references explicitly; a referee will otherwise do it for you.

## 6.4 Dicke-model engineering and Rydberg cavity QED — the feasibility base

The engineered Dicke model is experimentally established: proposed by Dimer *et al.* [11], realised by Baden *et al.* [12] with cavity-assisted Raman transitions, and separately by Baumann *et al.* [13] via self-organisation of a BEC. Cavity Rydberg polaritons have been observed [23], including the suppression of inhomogeneous broadening by collective strong coupling that this proposal needs. On the sensing side, Rydberg electrometers now approach the standard quantum limit in cold atoms [24] and reach within 1.0 dB of it in tweezer arrays [25], and collective-radiative linewidth engineering for Rydberg microwave sensing has just been proposed [26]. Rydberg systems can also be spin-squeezed directly [27], which is the main competing route to sub-SQL electrometry.

## 6.5 What appears to be unoccupied

On this survey, the following combination does not appear in the literature:

1. Using the *Dicke* (light-matter) superradiant transition, rather than a Rydberg-interaction transition, as the critical resource for microwave electrometry.

2. The *null / threshold-crossing* read-out of Protocol B, in which the measurand is the laser detuning that holds the system at threshold — SI-traceable, and with a discriminant not limited by the EIT linewidth.
3. The explicit gain–bandwidth characterisation of a critical Rydberg sensor, including the finding that there is a wide window of gain before critical slowing down is paid for.

Item 2 is the strongest. Items 1 and 3 are supporting.

## 7 Assessment

### 7.1 Is it a good idea?

**Yes, with one large caveat about how it is framed.** The physics is sound, the mapping from measurand to critical point is exact and simple, the required experimental ingredients all exist separately, and the numbers are competitive. But the natural framing — “criticality gives super-classical scaling for electrometry” — is not supportable, and Eq. (15) shows why in this specific system rather than by appeal to general theorems. The idea should be framed as *a tunable, SI-traceable critical amplifier for Rydberg electrometry*, and it is a good idea under that framing.

### 7.2 Benefits

- **Transduction gain where it is needed.**  $\varepsilon^{-1}$  gain, with roughly three decades of it free of bandwidth cost. Real Rydberg sensors are usually limited by optical detection and technical noise, not by atomic projection noise, so gain at the atomic stage is directly valuable.
- **In-situ tunable critical point.** Unlike the density-driven critical point of Refs. [6, 7],  $\lambda_c$  here is set by laser power and can be servoed, making the operating point stable and reproducible.
- **SI traceability via Protocol B.** Reading the measurand as a laser detuning preserves the metrological virtue that makes Rydberg electrometry attractive to national metrology institutes.
- **Self-generated squeezing.** Up to 10 dB, with no external squeezed source and no separate entanglement-generation step.
- **Bright-to-dark operation.** The response sign is favourable for detection.
- **A trigger mode.** With strong Rydberg interactions the transition can be made first order, giving a latching threshold detector.

### 7.3 Risks and objections

- **The scaling advantage is illusory.** Established in Sec. 4.3 and by Refs. [19, 20, 22]. This is the objection a referee will raise first.
- **Criticality amplifies drift as well as signal.** Protocol A has no discrimination between  $\delta\tilde{\omega}_0$  from the microwave field and  $\delta\lambda$  from laser intensity noise or  $\delta\omega$  from cavity drift. Protocol C’s IF encoding and Protocol B’s null lock are the mitigations, and they are not optional.
- **Bandwidth.** 27.5 kHz at the parameters used, set by  $(\kappa + \gamma)/2$ . That is fine for metrology and useless for a communications receiver. A higher- $\kappa$  cavity trades gain for bandwidth in the usual way.
- **Rydberg blockade caps  $N_{\text{eff}}$ ,** which is what Eq. (15) actually depends on. This is in direct tension with the Ref. [2] physics, which *wants* strong interactions.

- **Engineering overhead.** Two Raman channels plus a high-finesse cavity plus (probably) cold atoms is a substantial step away from the vapour-cell simplicity that makes Rydberg electrometry commercially interesting.
- **Decoherence cuts off the critical scaling.** The divergence is truncated once the soft-mode frequency falls below  $\gamma$ , i.e. at  $\varepsilon_{\min} \sim (\gamma/\omega)^2$ ; with vapour-cell linewidths this bites early.
- **Prior art is close.** Reference [7] in particular will be cited against any claim of novelty that is not carefully worded.

#### 7.4 Recommended programme

1. **Theory paper, correctly framed.** Title along the lines of “Critical Dicke amplification for SI-traceable Rydberg electrometry”. Lead with Protocol B (null read-out) as the novel element and the gain–bandwidth analysis as the practical one. State the no-go results up front rather than being caught by them.
2. **Do the calculation the referee will ask for.** A full Ziv–Zakai or Bayesian treatment of Protocol B’s threshold-location precision, since the Cramér–Rao bound is the wrong tool for a null measurement, and a proper account of inhomogeneous coupling across the cavity mode.
3. **Cheap experimental precursor.** Protocol C in the *Tavis–Cummings* regime — no counter-rotating terms, no phase transition — is essentially Ref. [3]’s apparatus plus a microwave LO. It would demonstrate the superheterodyne-in-a-cavity gain and de-risk everything except the Dicke engineering.
4. **Only then** add the second Raman channel and go for the transition.

## A Numerical methods

All code is in `code/`; all figures are regenerated by `code/make_figs.py`.

**Exact diagonalisation.** Equation (4) is built on the product of the  $j = N/2$  spin manifold ( $N + 1$  states) and a truncated Fock space. Since  $\partial H / \partial \tilde{\omega}_0 = J_z$  is parity-even, only the even-parity sector contributes to  $F_Q$  and the calculation is restricted to it, which also removes the closing odd-parity gap from the eigensolver’s path. Photon truncation was verified at  $N = 200$ ,  $\lambda = \lambda_c$ :  $F_Q$  agrees to twelve digits between  $n_{\max} = 60$  and  $n_{\max} = 120$ ; production runs used  $n_{\max} = 100$ –140. The QFI is obtained from the fidelity susceptibility with a symmetric step  $h = 10^{-4} \tilde{\omega}_0$ .

**Quantum Langevin.** The linearised open-system calculation uses symmetrised two-sided spectra with vacuum inputs, diffusion matrix  $D = \text{diag}(\kappa, \kappa, \gamma, \gamma)$ , and the input–output relation  $x_{\text{out}} = \sqrt{2\kappa} x_a - x_{\text{in}}$  including the cross-correlation between intracavity field and input vacuum. The implementation returns exactly  $S_{\text{out}} = 1/2$  for an empty cavity, which fixes all normalisation conventions. The damped threshold  $\lambda_c$  is located by bisection on  $\max \text{Re eig}(A) = 0$ . Homodyne angles are optimised on a 361-point grid.

**Parameters.** Unless stated:  $\omega/2\pi = \tilde{\omega}_0/2\pi = 1$  MHz,  $\kappa/2\pi = 50$  kHz,  $\gamma/2\pi = 5$  kHz,  $d = 1774.82$  ea<sub>0</sub> for  $^{87}\text{Rb } 53D_{5/2} \rightarrow 54P_{3/2}$ .

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